LEVEL UP ACADEMY

Class 10 | GSEB | Science

Ch 10 Test (GSEB pattern): The Human Eye and the Colourful World

Chapter / Topic: The Human Eye and the Colourful World

Name: _____ Roll No.: _____ Date: _____

Time: 50 minutes **Max. Marks: 25**

General instructions:

1. All questions are compulsory.
2. Section A: objective questions, 1 mark each. Section B: 2 marks each. Section C: 3 marks each. Section D: 4 marks.
3. Write balanced equations and draw neat, labelled diagrams wherever needed.

SECTION A

9 marks

Choose the correct option.

Q1. The human eye can focus on objects at different distances by adjusting the focal length of the eye lens. This is due to: **[1]**

(a) presbyopia (b) near-sightedness
(c) far-sightedness (d) accommodation

Q2. Which part of the eye controls the size of the pupil? **[1]**

(a) Iris (b) Cornea
(c) Retina (d) Eye lens

Q3. The human eye forms the image of an object at its: **[1]**

(a) cornea (b) retina
(c) iris (d) pupil

Q4. The splitting of white light into its component colours is called: **[1]**

(a) reflection (b) scattering
(c) dispersion (d) refraction

Q5. The least distance of distinct vision for a young adult with normal vision is about: **[1]**

(a) 25 m (b) 2.5 cm
(c) 25 cm (d) 2.5 m

Q6. The clear sky looks blue because of: **[1]**

(a) scattering of light (b) reflection of light
(c) dispersion of light (d) atmospheric refraction

Q7. The change in focal length of the eye lens is caused by the action of the: **[1]**

(a) pupil (b) retina
(c) iris (d) ciliary muscles

Q8. A person cannot see objects beyond 2 m clearly. The defect is: **[1]**

(a) presbyopia (b) myopia
(c) hypermetropia (d) cataract

Q9. Myopia (near-sightedness) is corrected using a: **[1]**

(a) convex lens (b) concave lens
(c) bifocal lens (d) cylindrical lens

SECTION B

6 marks

Q10. Why do stars twinkle? **[2]**

Q11. What are the near point and the far point of a normal human eye? **[2]**

Q12. What is dispersion of white light? What causes it? **[2]**

SECTION C

6 marks

Q13. A student can see distant objects clearly but cannot read a book held at a normal distance. **[3]**

- **(i)** Name this defect of vision. **[1]**
- **(ii)** Give two possible causes. **[1]**
- **(iii)** Name the lens used to correct it, and say how it helps. **[1]**

Q14. A person needs a lens of power −5.5 D to correct distant vision and a lens of power +1.5 D to correct near vision. **[3]**

- **(i)** Find the focal length of the lens for distant vision. **[1]**
- **(ii)** Find the focal length of the lens for near vision. **[1]**
- **(iii)** What kind of spectacle lens would suit this person? **[1]**

SECTION D

4 marks

Read the passage and answer the questions that follow.

Q15. A rainbow is a natural spectrum that appears in the sky after a rain shower. It is caused by the dispersion of sunlight by tiny water droplets in the atmosphere. A rainbow always forms in the direction opposite to the Sun. The droplets act like small prisms: they refract and disperse the sunlight, reflect it internally, and refract it again as it comes out of the droplet. **[4]**

- **(i)** In which direction must you face to see a rainbow? **[1]**
- **(ii)** What acts as a prism in the formation of a rainbow? **[1]**
- **(iii)** Name, in order, the three phenomena of light that occur inside a raindrop. **[2]**

~ All the best! ~